Anatomical PDI registration

fUS_block_PDI_float.bin
binary pdi data

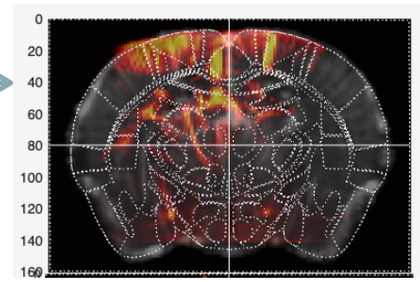
rawPDI

anatomic.mat

BFConfig

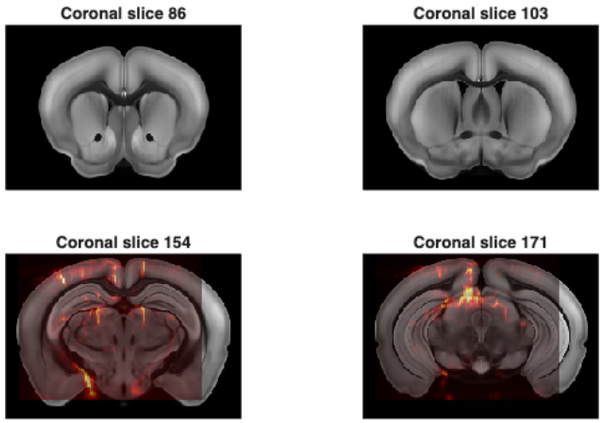
L22-14_PlaneWave_FUSI_data.mat
info about X,Y,t and dims

registration to allen
s01_Register_Anatomic_2_Atlas.m

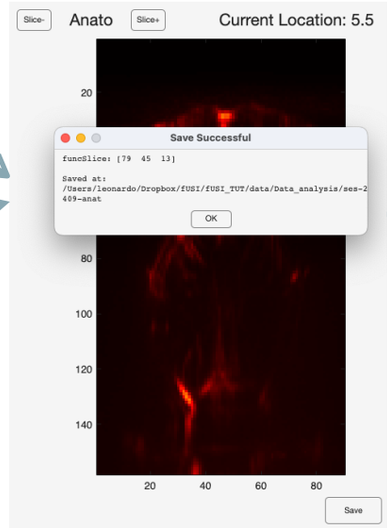


Transformation.mat
subject —> atlas

slice into atlas space
s03_Transform_Individual_2_Atlas



select slice
s02_Select_Anatomical_Slice.m



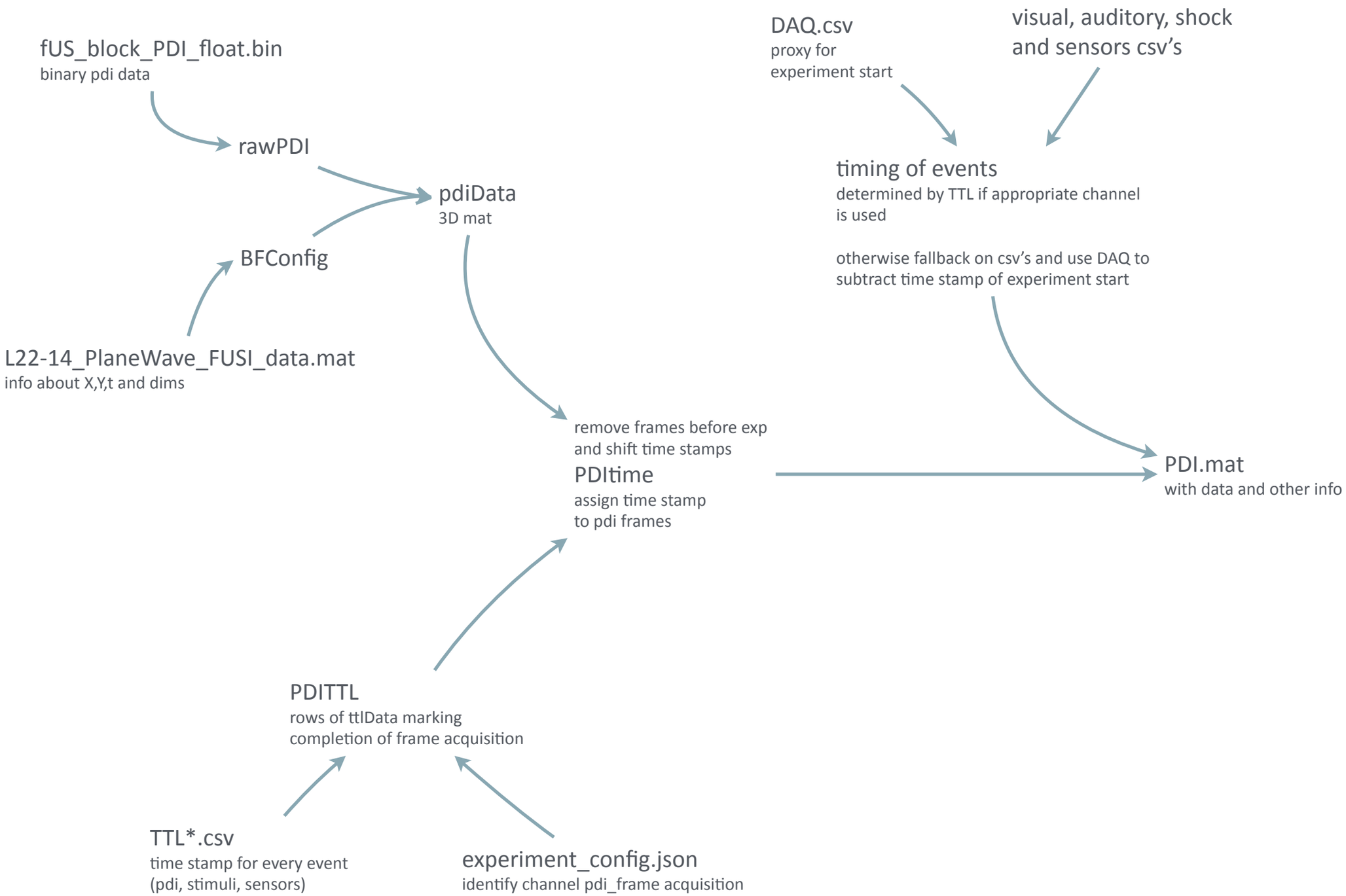
MotorScan.tsv

anatomic.mat
add field:
anatomic.funcSlice [79.45.13]

anatomic_2_atlas.mat
anatomic_2_atlas.nii
full anatomic acquisition in atlas space

slice_2_atlas.nii
slice_2_atlas_mask.nii
selected slice (and its mask) in atlas space.
useful to then select the regions from the allen atlas

Functional PDI reconstruction



DAQ.csv

	A	B	C	D	E	F	G	H	I
1	time	visual	audio	shock_left	shock_right	shock_tail	recording_start	stim_start	camera_ttl_trigger
2	1702637451.57563	0	0	1	1	1	0	0	0
3	1702637451.57563	0	0	1	1	1	0	0	0
4	1702637463.19438	0	0	1	1	1	0	1	1
5	1702637463.20535	0	0	1	1	1	0	0	1
6	1702637583.21739	1	0	1	1	1	0	0	1
7	1702637598.20012	0	0	1	1	1	0	0	1
8	1702637643.2082	1	0	1	1	1	0	0	1
9	1702637658.21268	0	0	1	1	1	0	0	1
10	1702637703.19373	1	0	1	1	1	0	0	1
11	1702637718.20773	0	0	1	1	1	0	0	1
12	1702637763.19493	1	0	1	1	1	0	0	1
13	1702637778.19716	0	0	1	1	1	0	0	1
14	1702637823.20051	1	0	1	1	1	0	0	1
15	1702637838.20534	0	0	1	1	1	0	0	1
16	1702637883.20701	1	0	1	1	1	0	0	1
17	1702637898.19484	0	0	1	1	1	0	0	1
18	1702637943.21178	1	0	1	1	1	0	0	1
19	1702637958.20005	0	0	1	1	1	0	0	1
20	1702638003.2109	1	0	1	1	1	0	0	1
21	1702638018.19991	0	0	1	1	1	0	0	1
22	1702638063.20713	1	0	1	1	1	0	0	1
23	1702638078.20913	0	0	1	1	1	0	0	1
24	1702638123.21645	1	0	1	1	1	0	0	1
25	1702638138.21145	0	0	1	1	1	0	0	1
26	1702638189.31299	0	0	1	1	1	0	0	0

TTL.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	0	0	1	1	1	0	0	0	0	0	0	1	0
2	0.0002	0	1	1	1	0	0	0	0	0	0	1	0
3	0.0004	0	1	1	1	0	0	0	0	0	0	1	0
4	0.0006	0	1	1	1	0	0	0	0	0	0	1	0
5	0.0008	0	1	1	1	0	0	0	0	0	0	1	0
6	0.001	0	1	1	1	0	0	0	0	0	0	1	0
7	0.0012	0	1	1	1	0	0	0	0	0	0	1	0
8	0.0014	0	1	1	1	0	0	0	0	0	0	1	0
9	0.0016	0	1	1	1	0	0	0	0	0	0	1	0
10	0.0018	0	1	1	1	0	0	0	0	0	0	1	0

VisualStimulation.csv

A	B	C	D
time	stim	event	duration
1702637463.19039	visual	start	0
1702637583.20343	visual	stim	15
1702637598.19812	visual	blank	0
1702637643.20521	visual	stim	15
1702637658.20271	visual	blank	0
1702637703.19074	visual	stim	15
1702637718.20474	visual	blank	0
1702637763.19294	visual	stim	15
1702637778.19317	visual	blank	0
1702637823.19552	visual	stim	15
1702637838.19536	visual	blank	0

DAQ.csv

time	visual	audio	shock_left	shock_right	shock_tail	recording_start	stim_start	camera_ttl_trigger
1702637451.57563	0	0	1	1	1	0	0	0
1702637451.57563	0	0	1	1	1	0	0	0
1702637463.19438	0	0	1	1	1	0	1	1
1702637463.20535	0	0	1	1	1	0	0	1
1702637583.21739	1	0	1	1	1	0	0	1
1702637598.20012	0	0	1	1	1	0	0	1
1702637643.2082	1	0	1	1	1	0	0	1
1702637658.21268	0	0	1	1	1	0	0	1
1702637703.19373	1	0	1	1	1	0	0	1
1702637718.20773	0	0	1	1	1	0	0	1
1702637763.19493	1	0	1	1	1	0	0	1
1702637778.19716	0	0	1	1	1	0	0	1
1702637823.20051	1	0	1	1	1	0	0	1
1702637838.20534	0	0	1	1	1	0	0	1
1702637883.20701	1	0	1	1	1	0	0	1
1702637898.19484	0	0	1	1	1	0	0	1
1702637943.21178	1	0	1	1	1	0	0	1
1702637958.20005	0	0	1	1	1	0	0	1
1702638003.2109	1	0	1	1	1	0	0	1
1702638018.19991	0	0	1	1	1	0	0	1
1702638063.20713	1	0	1	1	1	0	0	1
1702638078.20913	0	0	1	1	1	0	0	1
1702638123.21645	1	0	1	1	1	0	0	1
1702638138.21145	0	0	1	1	1	0	0	1
1702638189.31299	0	0	1	1	1	0	0	0

TTL.csv

Determining Experiment start

Both the pdiData and the time stamp of pdi frames and other events (e.g. stimuli) is set to the beginning of the experiment (experiment start). This is done according to the info in the appropriate channel in TTL.

TTL also records the timing of the stimuli and other events (e.g. wheel) in appropriate channels, however *in some cases* this info might be missing from the TTL. In these cases, the first record of the **DAQ.csv** file is taken as experiment start. This is then removed from the csv of the single instruments, e.g. VisualStimulation.csv

VisualStimulation.csv

A	B	C	D
time	stim	event	duration
1702637463.19039	visual	start	0
1702637583.20343	visual	stim	15
1702637598.19812	visual	blank	0
1702637643.20521	visual	stim	15
1702637658.20271	visual	blank	0
1702637703.19074	visual	stim	15
1702637718.20474	visual	blank	0
1702637763.19294	visual	stim	15
1702637778.19317	visual	blank	0
1702637823.19552	visual	stim	15
1702637838.19536	visual	blank	0

DAQ.csv

time	visual	audio	shock_left	shock_right	shock_tail	recording_start	stim_start	camera_ttl_trigger
1702637451.57563	0	0	1	1	1	0	0	0
1702637451.57563	0	0	1	1	1	0	0	0
1702637463.19438	0	0	1	1	1	0	1	1
1702637463.20535	0	0	1	1	1	0	0	1
1702637583.21739	1	0	1	1	1	0	0	1
1702637598.20012	0	0	1	1	1	0	0	1
1702637643.2082	1	0	1	1	1	0	0	1
1702637658.21268	0	0	1	1	1	0	0	1
1702637703.19373	1	0	1	1	1	0	0	1
1702637718.20773	0	0	1	1	1	0	0	1
1702637763.19493	1	0	1	1	1	0	0	1
1702637778.19716	0	0	1	1	1	0	0	1
1702637823.20051	1	0	1	1	1	0	0	1
1702637838.20534	0	0	1	1	1	0	0	1
1702637883.20701	1	0	1	1	1	0	0	1
1702637898.19484	0	0	1	1	1	0	0	1
1702637943.21178	1	0	1	1	1	0	0	1
1702637958.20005	0	0	1	1	1	0	0	1
1702638003.2109	1	0	1	1	1	0	0	1
1702638018.19991	0	0	1	1	1	0	0	1
1702638063.20713	1	0	1	1	1	0	0	1
1702638078.20913	0	0	1	1	1	0	0	1
1702638123.21645	1	0	1	1	1	0	0	1
1702638138.21145	0	0	1	1	1	0	0	1
1702638189.31299	0	0	1	1	1	0	0	0